

Name: Richard

Hammack

Score: _____

Directions: Do problems 1 and 3, and **one** of problems 2 or 4.

1. List the elements of the subgroup of $GL_2(\mathbb{R})$ generated by $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = A$

$$A^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$A^3 = A^2 A = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$A^4 = A^3 A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$\text{Therefore } \langle \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \rangle = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\}$$

2. Let p be a prime and take $r \in \mathbb{N}$. How many generators does $\mathbb{Z}_{p^r}$ have? Explain.

Notice that $a \in \mathbb{Z}_{p^r}$ is not a generator precisely when $\gcd(p^r, a) \neq 1$, that is, when a is a multiple of p .

Thus the elements of $\mathbb{Z}_{p^r}$ that are not generators are:

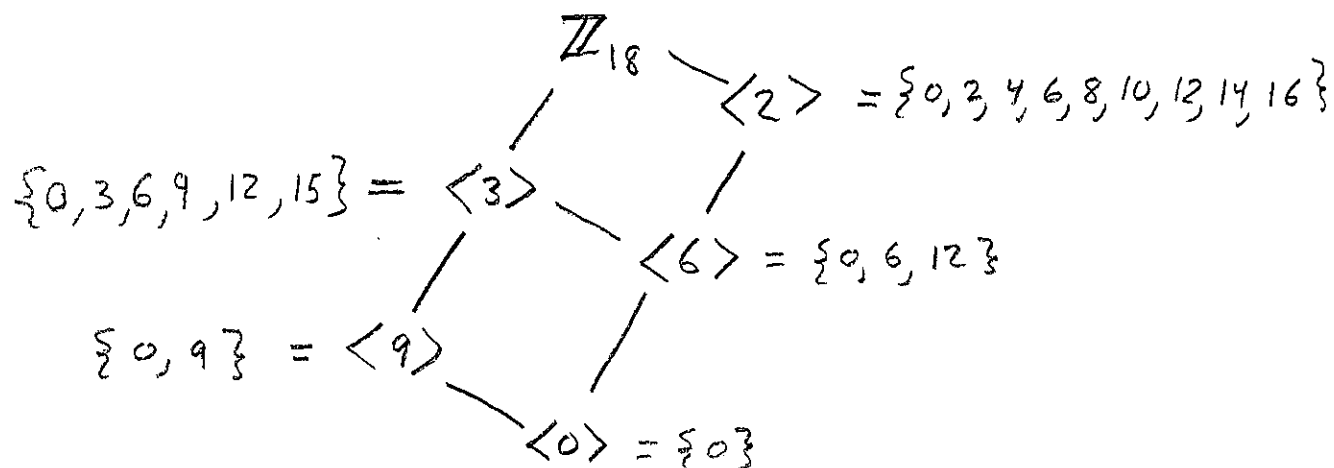
$$\underline{1} \cdot p, \underline{2} \cdot p, \underline{3} \cdot p, \underline{4} \cdot p, \dots, \underline{p^{r-1}} \cdot p = 0$$

Thus p^{r-1} elements of $\mathbb{Z}_{p^r}$ are not generators.

So the number of generators of $\mathbb{Z}_{p^r}$ is

$$p^r - p^{r-1} = \boxed{p^{r-1}(p-1)}.$$

3. List all subgroups of $\mathbb{Z}_{18}$. Consider arranging them in a subgroup lattice.



4. Suppose $a, b \in G$, for some group G . Prove that ab and ba have the same order.

We will prove that for any $k \in \mathbb{Z}$,
 $(ab)^k = e$ if and only if $(ba)^k = e$.

This will imply ab and ba have the same order, for then the smallest non-negative integer k for which $(ab)^k = e$ equals the smallest non-negative power for which $(ba)^k = e$.

$$\begin{aligned}
 \text{Observe: } (ab)^k = e &\iff b(ab)^k a = ba \\
 &\iff (ba)^{k+1} = ba \\
 &\iff (ba)^{k+1} (ba)^{-1} = (ba)(ba)^{-1} \\
 &\iff (ba)^k = e,
 \end{aligned}$$

as desired $\blacksquare$